智慧藥盒

結合處方辨識與自動分配的 IoT 智慧藥物管理系統

Smart IoT Pill Dispenser

Prescription-Aware Automated Medication Management System

Project Proposal

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Advisor:

Date: 17 September 2025

Project Proposal

智慧藥盒：處方辨識與自動分配系統

(Smart Pill Dispenser: Prescription-Aware Automated Medication System for Taiwan's Aging Society)

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1. Introduction & Problem Statement

By 2025, Taiwan becomes a super-aged society (20%+ population over 65). Elderly patients often juggle multiple prescriptions, which creates risks of:

- Medication errors: wrong pill, wrong time, overdose, or missed doses.
- Caregiver burden: family members must prepare daily pillboxes and remind elders.
- Health risks: poor adherence leads to preventable hospitalizations.

Current gap: Existing dispensers only remind users to take pills but cannot verify pill type, automatically sort pills, or adapt to Taiwan's NHI prescription system.

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2. Project Objectives

We propose to design an IoT-enabled smart pill dispenser with these goals:

1. Prescription-Aware Automation

- Import doctor's prescription via NHI QR code / app screenshot or scan/upload prescription paper.
- Parse medication names, dosages, and schedules automatically.

2. Pill Recognition & Sorting (Package-Based Intake)

- Users load one entire package of pills at a time into the container.
- System uses camera + ML to recognize each pill (shape, color, imprint).
- Pills are then sorted into correct compartments according to prescription.

3. Automated Dispensing

- At scheduled times, the device releases the correct set of pills into a cup.
- Provides voice, vibration, and LINE alerts as reminders.

4. Adherence Monitoring

- Detects pill retrieval (via load cell/sensor).
- Logs medication history and alerts caregivers for missed doses.

5. Lightweight Companion App

To complement the dispenser, we propose a simple companion app for caregivers and family members. The app remains lightweight and focused on usability rather than technical complexity. Core features include:

- **Prescription Import:** Scan/upload NHI QR code or photo of doctor's prescription.
- **Schedule Overview:** View daily/weekly medication schedule in Traditional Chinese.
- **Reminders & Alerts:** Receive notifications of taken/missed doses (mirrored to LINE).
- **Adherence Tracking:** Access history of medication intake, with missed-dose logs.
- **Caregiver Integration:** Allow family members to monitor and receive alerts remotely.
- **Language Support:** Traditional Chinese interface with Mandarin/Hokkien voice prompt options.
- **Simple Setup:** Pair dispenser, adjust time settings, and manage basic configurations.

The app is intentionally minimal: elderly users interact primarily with the device and LINE notifications, while caregivers use the app to set up, monitor, and adjust prescriptions when needed.

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3. Novelty & Contribution

- End-to-End Integration: Combines prescription parsing, pill recognition, sorting, and dispensing.
- Taiwan Localization:
  - Traditional Chinese interface, Mandarin/Hokkien voice prompts.
  - LINE notification system.
  - Support for NHI prescription QR codes.
- Affordable Modular Design: Core dispenser with optional caregiver app.
- Social Impact: Supports Long-Term Care 2.0 and Smart Healthcare initiatives.

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4. Technical Approach

Hardware

- Microcontroller: ESP32 / Raspberry Pi
- Sensors: PiCam (pill recognition), load cell (dispense confirmation)
- Motors: Stepper/servo for sorting & dispensing
- Connectivity: WiFi/GSM for LINE integration
- Output: Speaker, LED, vibration motor

Software

- Prescription parser: QR/barcode → structured schedule & pill list
- Computer Vision: TensorFlow Lite model for pill recognition
- IoT Platform: MQTT or Firebase + LINE API integration
- Scheduler: Embedded logic for dispensing + adherence monitoring

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5. Development Plan & Timeline (Two Semesters)

Semester 1 (Sep 2025 – Jan 2026)

- Prescription parsing (NHI QR code, app input)
- Hardware setup + basic pill container (package intake)
- Pill recognition prototype with ML model
- Sorting mechanism: package → compartments
- Mid-term demo: package intake → pill recognition → sorting

Semester 2 (Feb 2026 – Jun 2026)

- Automated dispensing module (compartment → cup)
- Reminders (voice, vibration, LINE integration)
- Adherence monitoring (sensors + caregiver alerts)
- Usability testing with elderly/caregiver groups
- Final refinement, UI/UX polish, bilingual prompts
- Final demo + presentation

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6. Expected Outcomes

- A working prototype:
  - Intake: loads one pill package at a time.
  - Sorting: places pills into correct compartments.
  - Dispensing: releases pills on schedule.
  - Alerts: LINE, vibration, and voice.
- Demonstration with 3–5 pill types and sample Taiwan prescriptions.
- Evaluation of recognition accuracy, user feedback, and adherence improvement.

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7. Impact

- Academic: Combines IoT, ML, and healthcare technologies.
- Social: Reduces caregiver stress and helps elders maintain independence.
- Practical: Ready for real-world testing in clinics, pharmacies, and care centers.

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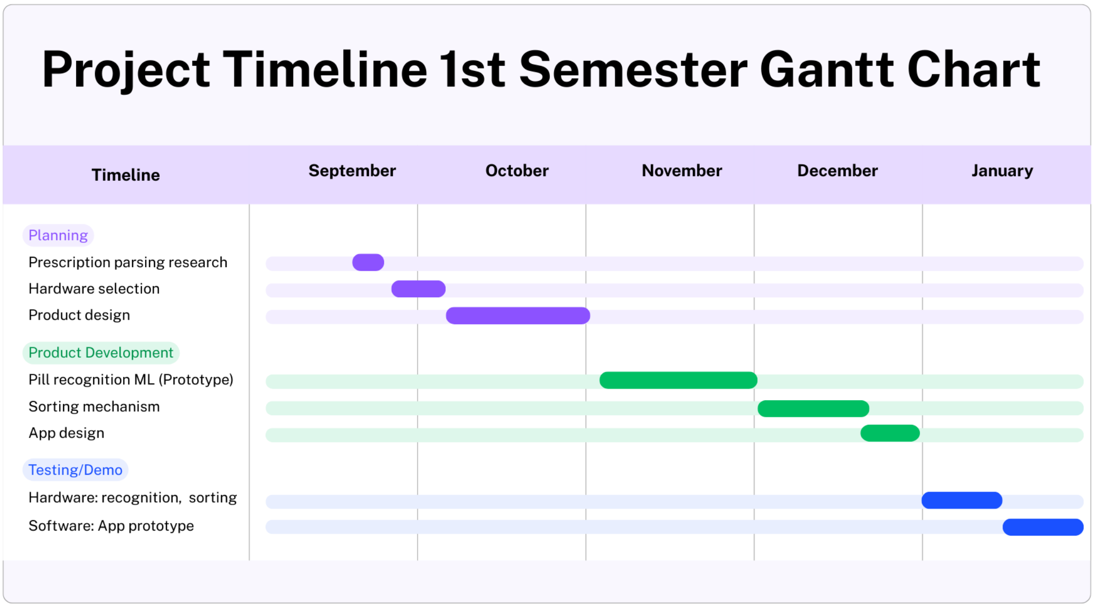
8. Conclusion

This project creates a next-generation smart pill dispenser for Taiwan's aging society. By integrating AI-driven pill recognition, NHI prescription parsing, and IoT-enabled reminders, it addresses a critical healthcare gap and offers a localized, affordable solution for elderly medication management.

Gantt-Style Timeline (Sep 2025 – Jun 2026)

Semester 1 (Fall 2025)

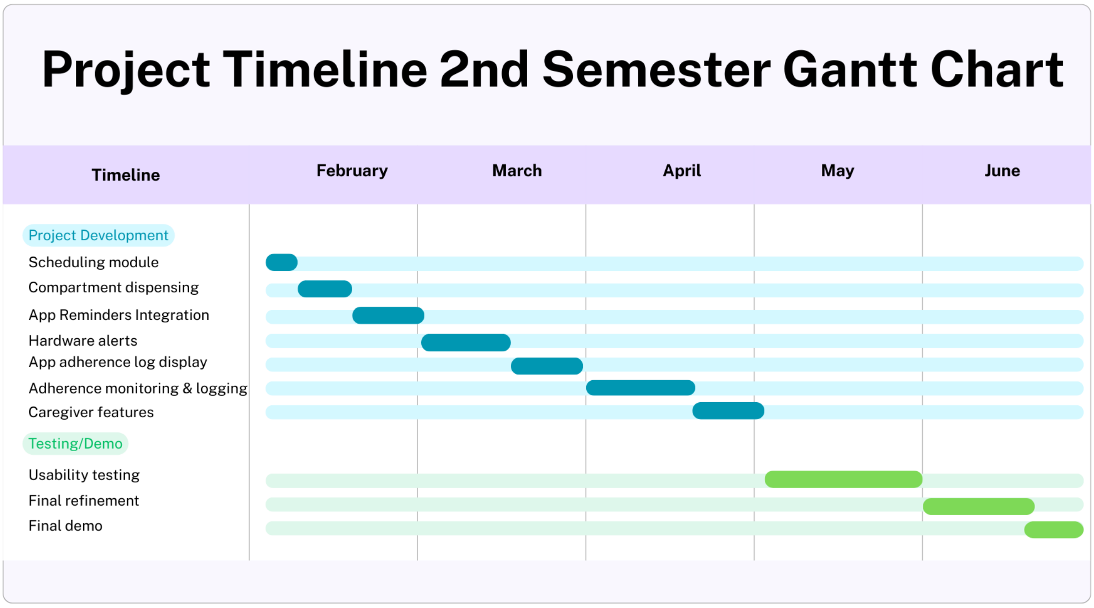


Month-by-month plan

- **Sep** — Prescription parsing, hardware selection
- **Oct** — Basic pill container & package intake design
- **Nov** — Pill recognition ML (prototype dataset)
- **Dec** — Sorting mechanism (package → compartments), app wireframes & design
- **Jan** — Mid-term demo (intake + recognition + sorting), app prototype (prescription import + schedule view)

Gantt-Style Timeline (Sep 2025 – Jun 2026)

Semester 2 (Spring 2026)



Month-by-month plan

- **Feb** — Scheduling module + compartment dispensing; app reminders integration
- **Mar** — LINE reminders + hardware alerts; app adherence log display
- **Apr** — Adherence monitoring + logging; caregiver features (multi-user alerts)
- **May** — Usability testing (elderly/caregivers) for dispenser & app
- **Jun** — Final refinement + bilingual UI + final demo (device + companion app)